CAR DEKHO PROJECT

1. Create database and then create table.

Code:

```sql
CREATE TABLE car_dekho (
    name text,
    year INT8,
    selling_price INT8,
    km_driven INT8,
    fuel VARCHAR(50),
    seller_type VARCHAR(50),
    transmission VARCHAR(50),
    owner VARCHAR(50),
    mileage VARCHAR(50),
    engine VARCHAR(50),
    max_power VARCHAR(50),
    torque VARCHAR(50),
    seats INT8
);
```

2. Import csv files:

Code:

```sql
copy car_dekho(name, year, selling_price, km_driven, fuel, seller_type, transmission, owner, mileage, engine, max_power, torque,seats)
From 'S:\Project Data analysis\SQL Project\Car_dekho.csv'
delimiter ','
csv header;
```

3. check if dataset imported successfully?

Code:

```sql
select * from car_dekho;
```

Ouput:

Data Output | Messages | Graph Visualiser | Notifications

Showing rows: 1 to 1000 Page No: 1 of 17

	name text	year bigint	selling_price bigint	km_driven bigint	fuel character varying (50)	seller_type character varying (50)	transmission character varying (50)	owner character varying (50)	mileage character varying (50)	engine character varying (50)	max_power character varying (50)	torque character varying (50)	seats bigint
1	Maruti Alto 800 LXI Opt	2023	410000	10000	Petrol	Individual	Manual	First Owner	19.03 kmpl	999 CC	71.01bhp	96Nm	5
2	Skoda Slavia 1.0 TSI Ambition	2023	1350000	10000	Petrol	Individual	Manual	First Owner	14.08 kmpl	1956 CC	167.67bhp	350nm	5
3	BMW 3 Series Gran Limousine 320Ld Luxury Line	2023	5800000	1000	Diesel	Dealer	Automatic	First Owner	18.15 kmpl	998 CC	118.35bhp	172Nm	5
4	MG ZS EV Exclusive	2023	2650000	10000	Electric	Dealer	Automatic	First Owner	32.52 kmpl	998 CC	58.33bhp	78Nm	5
5	Tata Punch Adventure	2023	715000	10000	Petrol	Individual	Manual	First Owner	12.15 kmpl	1451 CC	141bhp	250Nm	5
6	Maruti S-Presso VXi Plus	2023	450000	30171	Petrol	Individual	Manual	First Owner	19.03 kmpl	999 CC	71.01bhp	96Nm	5
7	Maruti S-Presso LXi	2022	425000	1994	Petrol	Dealer	Manual	First Owner	19.47 kmpl	999 CC	113.98bhp	178Nm	5
8	Hyundai Creta SX Turbo	2022	1895000	22000	Petrol	Individual	Automatic	First Owner	12.15 kmpl	1997 CC	296.3bhp	400Nm	5
9	Renault Kiger RXT AMT Opt DT	2022	842000	6424	Petrol	Individual	Automatic	First Owner	14.08 kmpl	1956 CC	167.67bhp	350nm	5
10	Renault KWID CLIMBER	2022	567000	5148	Petrol	Dealer	Manual	First Owner	18.15 kmpl	998 CC	118.35bhp	172Nm	5
11	Mahindra XUV300 W8 Diesel Sunroof	2022	1197000	5030	Diesel	Individual	Manual	Second Owner	32.52 kmpl	998 CC	58.33bhp	78Nm	5
12	Mahindra XUV700 AX5 Diesel AT	2022	2275000	28000	Diesel	Individual	Automatic	First Owner	12.15 kmpl	1451 CC	141bhp	250Nm	5
13	Renault Triber RXT	2022	800000	10000	Petrol	Individual	Manual	Second Owner	21.01 kmpl	1197 CC	81.80bhp	113Nm	5
14	Hyundai Creta SX Diesel AT	2021	1950000	8000	Diesel	Individual	Automatic	First Owner	13.38 kmpl	2993 CC	265 bhp	620Nm@ 1500-2500rpm	5
15	Renault KWID CLIMBER	2021	567000	5868	Petrol	Individual	Manual	First Owner	18.15 kmpl	998 CC	118.35bhp	172Nm	5
16	Nissan Magnite XV Premium	2021	850000	2963	Petrol	Individual	Manual	First Owner	32.52 kmpl	998 CC	58.33bhp	78Nm	5
17	Renault Triber RXZ	2021	785000	11192	Petrol	Individual	Manual	Second Owner	12.15 kmpl	1451 CC	141bhp	250Nm	5
18	Hyundai Tucson Platinum AT	2021	2975000	3600	Petrol	Dealer	Automatic	First Owner	21.01 kmpl	1197 CC	81.80bhp	113Nm	5
19	Maruti Alto 800 LXI Opt	2021	358000	29751	Petrol	Dealer	Manual	First Owner	19.03 kmpl	999 CC	71.01bhp	96Nm	5
20	Hyundai i20 Sportz Diesel	2021	977000	13522	Diesel	Individual	Manual	First Owner	18.15 kmpl	2198 CC	182.38bhp	450Nm	5
21	BMW X7 xDrive 30d DPE	2020	7200000	5000	Diesel	Individual	Automatic	First Owner	13.38 kmpl	2993 CC	265 bhp	620Nm@ 1500-2500rpm	7
22	Datsun GO T Option	2020	450000	10000	Petrol	Individual	Manual	First Owner	19.02 kmpl	1198 CC	67.05 bhp	104Nm@ 4000rpm	5
23	Datsun RediGO 1.0 S	2020	350000	2136	Petrol	Individual	Manual	First Owner	22.5 kmpl	999 CC	67 bhp	91Nm@ 4250rpm	5
24	Ford Freestyle Titanium Petrol BSIV	2020	400000	5000	Petrol	Individual	Manual	Second Owner	19.0 kmpl	1194 CC	94.68 bhp	120Nm@ 4250rpm	5
25	Honda Civic ZX Diesel BSIV	2020	2125000	5500	Diesel	Dealer	Manual	First Owner	26.8 kmpl	1597 CC	118 bhp	300Nm@ 2000rpm	5
26	Honda Civic ZX Diesel BSIV	2020	2125000	5500	Diesel	Dealer	Manual	First Owner	26.8 kmpl	1597 CC	118 bhp	300Nm@ 2000rpm	5
27	Hyundai Creta 1.4 EX Diesel	2020	1050000	20000	Diesel	Individual	Manual	First Owner	22.1 kmpl	1396 CC	88.7 bhp	219.7Nm@ 1500-2750rpm	5
28	Hyundai Creta 1.6 CRDi AT SX Plus	2020	1950000	2700	Diesel	Dealer	Automatic	First Owner	17.01 kmpl	1582 CC	126.2 bhp	259.9Nm@ 1900-2750rpm	5
29	Hyundai Elite i20 Sportz Plus BSIV	2020	720000	5000	Petrol	Individual	Manual	First Owner	18.6 kmpl	1197 CC	81.86 bhp	117nm@ 4000rpm	5
30	Hyundai Grand i10 Nios AMT Sportz	2020	675000	5000	Petrol	Individual	Automatic	First Owner	20.7 kmpl	1197 CC	81.86 bhp	113.75Nm@ 4000rpm	5
31	Hyundai Grand i10 Nios Sportz	2020	775000	2000	Petrol	Dealer	Manual	First Owner	20.7 kmpl	1197 CC	81.86 bhp	113.75Nm@ 4000rpm	5
32	Hyundai Grand i10 Sportz	2020	330000	70000	Petrol	Individual	Manual	Second Owner	18.9 kmpl	1197 CC	81.86 bhp	113.75nm@ 4000rpm	5
33	Hyundai Santro Magna	2020	500000	1000	Petrol	Individual	Manual	First Owner	20.3 kmpl	1086 CC	68.07 bhp	99.07Nm@ 4500rpm	5
34	Hyundai Xcent 1.2 VTVT E	2020	360000	65000	Petrol	Individual	Manual	Second Owner	20.14 kmpl	1197 CC	81.86 bhp	113.75nm@ 4000rpm	5
35	Hyundai Xcent 1.2 VTVT SX	2020	524000	32000	Petrol	Individual	Manual	First Owner	20.14 kmpl	1197 CC	81.86 bhp	113.75nm@ 4000rpm	5
36	Hyundai Xcent 1.2 VTVT SX	2020	524000	32000	Petrol	Individual	Manual	First Owner	20.14 kmpl	1197 CC	81.86 bhp	113.75nm@ 4000rpm	5
37	Kia Seltos HTX Plus AT D	2020	1900000	5000	Diesel	Dealer	Automatic	First Owner	17.8 kmpl	1493 CC	113.4 bhp	250nm@ 1500-2750rpm	5
38	Kia Seltos HTX Plus AT D	2020	1900000	5000	Diesel	Dealer	Automatic	First Owner	17.8 kmpl	1493 CC	113.4 bhp	250nm@ 1500-2750rpm	5

Total rows: 16296 | Query complete 00:00:00.429 CRLF Ln 36, Col 1

4. **Total Cars: To get a count of total records:**

Code:

```
select count(*) from car_dekho;
```

Output:

	count bigint
1	16296

5. **How many cars will be available in 2023?**

Code:

```
select count(*)
from car_dekho
where year = 2023;
```

Output:

	count bigint
1	12

6. **How many cars is available in 2020, 2021, 2022?**

Code:

```
select year, count(*)
from car_dekho
where year in (2020, 2021, 2022)
group by year;
```

Output:

	year bigint	count bigint
1	2022	14
2	2020	148
3	2021	14

7. **How many diesel cars will there be in 2020?**

Code:

```
select count(*)
from car_dekho
where fuel = 'Diesel' and year=2020;
```

Output:

	count bigint
1	40

8. **Print all the fuel cars (petrol, diesel, and CNG) come by all year.**

Code:

```
SELECT
    year,
    COUNT(*) FILTER (WHERE fuel = 'Petrol') AS petrol_count,
    COUNT(*) FILTER (WHERE fuel = 'Diesel') AS diesel_count,
    COUNT(*) FILTER (WHERE fuel = 'CNG') AS cng_count
FROM car_dekho
GROUP BY year
ORDER BY year;
```

Output:

	year bigint	petrol_count bigint	diesel_count bigint	cng_count bigint
1	1983	0	2	0
2	1991	0	0	0
3	1994	2	4	0
4	1995	2	2	0
5	1996	4	2	0
6	1997	18	4	0
7	1998	18	2	0
8	1999	26	8	0
9	2000	28	14	2
10	2001	20	0	0
11	2002	46	8	0
12	2003	70	26	2
13	2004	100	24	0
14	2005	138	56	0
15	2006	178	68	0
16	2007	208	146	4
17	2008	228	190	2
18	2009	282	210	0
19	2010	384	380	6
20	2011	424	742	10
21	2012	404	868	10
22	2013	406	924	6
23	2014	404	830	8
24	2015	556	988	4
25	2016	858	848	12
26	2017	864	1154	18
27	2018	788	816	10
28	2019	704	448	14
29	2020	102	40	6
30	2021	10	4	0
31	2022	10	4	0
32	2023	8	2	0

9. **There were more than 100 cars in a given year, which year had more than 100 cars?**

Code:

```
SELECT
    year,
    COUNT(*) AS total_cars
FROM car_dekho
GROUP BY YEAR
HAVING COUNT(*) > 100
ORDER BY COUNT(*) DESC;
```

Output:

	year bigint	total_cars bigint
1	2017	2036
2	2016	1718
3	2018	1614
4	2015	1552
5	2013	1340
6	2012	1302
7	2014	1242
8	2011	1184
9	2019	1166
10	2010	788
11	2009	492
12	2008	428
13	2007	366
14	2006	248
15	2005	194
16	2020	148
17	2004	124

10. **All cars count details between 2015 and 2023.**

Code:

```
SELECT
    year,
    COUNT(*)
FROM car_dekho
WHERE year BETWEEN 2015 AND 2023
GROUP BY YEAR
ORDER BY YEAR ASC;
```

Output:

	year bigint	count bigint
1	2015	1552
2	2016	1718
3	2017	2036
4	2018	1614
5	2019	1166
6	2020	148
7	2021	14
8	2022	14
9	2023	12